DALNERECHENSK, Russian Federation

WMO: **318730**

Lat: **45.8796N** Lon: **133.7245E** Elev: **105** StdP: **100.08** Time Zone: **10.00 (E10)** Period: **94-19** WBAN: **99999**

Annual Heating, Humidification, and Ventilation Design Conditions

	Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF	
				99.6%			99%			0.4%		1%					
		99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD		
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(1)	1	-27.8	-26.0	-31.5	0.2	-26.7	-29.9	0.2	-24.9	7.2	-13.7	6.4	-13.9	1.8	260	0.519	(1)

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

	Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB		
			0.4%		1%		2%		0.4%		1%		2%				
			DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD	
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(2)	7	8.0	29.1	22.3	27.8	21.5	26.5	20.7	23.7	27.3	22.8	26.2	21.9	25.1	2.8	230	(2)

	Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB	
	0.4%			1%			2%			0.4%		1%		2%			
	DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB		
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(3)	22.6	17.5	25.8	21.7	16.5	24.9	20.8	15.6	24.1	72.1	27.5	68.3	26.3	65.0	25.2	27.4	(3)

Extreme Annual Design Conditions

	Extreme Annual WS				Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
					Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
	1%	2.5%	5%		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
	(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(4)	7.6	6.5	5.7	DB	-30.5	31.7	2.7	2.1	-32.5	33.2	-34.0	34.4	-35.5	35.6	-37.5	37.1	(4)
(5)				WB	-30.7	25.3	2.7	1.2	-32.6	26.1	-34.2	26.8	-35.7	27.5	-37.6	28.4	(5)

Monthly Climatic Design Conditions

				Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
				(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg		3.8	-18.2	-13.5	-4.7	5.9	13.5	18.5	21.8	20.7	14.7	6.3	-5.4	-15.7	(6)
(7)		DBStd		14.74	4.48	5.00	5.65	4.44	3.69	3.16	2.38	2.70	3.49	4.71	6.50	5.05	(7)
(8)		HDD10.0		3531	873	658	455	137	10	0	0	0	4	132	463	798	(8)
(9)		HDD18.3		5569	1132	892	713	372	156	36	3	9	116	372	712	1056	(9)
(10)		CDD10.0		1252	0	0	0	15	118	254	367	333	146	18	0	0	(10)
(11)		CDD18.3		249	0	0	0	0	5	41	112	84	8	0	0	0	(11)
(12)		CDH23.3		1422	0	0	0	3	57	266	636	419	41	1	0	0	(12)
(13)		CDH26.7		255	0	0	0	0	7	48	136	62	3	0	0	0	(13)
(14)	Wind	WSAvg		2.6	2.5	2.7	3.0	3.2	3.0	2.4	2.2	2.1	2.3	2.9	2.9	2.5	(14)
(15)	Precipitation	PrecAvg		637	12	11	18	43	57	84	112	120	81	52	27	17	(15)
(16)		PrecMax		907	34	41	48	133	129	191	271	267	190	118	63	38	(16)
(17)		PrecMin		369	0	0	0	9	2	8	21	19	12	0	8	3	(17)
(18)		PrecStd		111	8	9	12	24	30	44	59	66	41	32	14	9	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB		-2.6	3.2	11.2	22.9	27.8	30.1	31.4	29.8	26.7	21.5	13.0	2.3	(19)
(20)			MCWB		-3.9	0.0	5.0	12.8	17.2	21.5	23.9	23.9	19.9	13.7	8.2	0.9	(20)
(21)		2%	DB		-6.8	0.5	8.0	19.6	25.0	27.9	29.6	28.4	24.4	18.6	9.2	-2.0	(21)
(22)			MCWB		-8.0	-2.1	3.0	10.7	15.7	19.9	23.0	22.6	18.4	12.0	5.6	-3.3	(22)
(23)		5%	DB		-9.4	-2.6	5.6	16.4	22.8	26.3	28.1	27.1	22.6	16.0	6.4	-5.1	(23)
(24)			MCWB		-10.6	-4.6	1.5	8.9	14.4	19.4	22.2	21.7	17.3	10.4	3.5	-6.3	(24)
(25)		10%	DB		-11.5	-5.5	3.4	13.7	20.7	24.6	26.8	25.7	21.0	13.9	3.7	-8.3	(25)
(26)			MCWB		-12.5	-7.2	0.2	7.3	13.2	18.5	21.4	20.8	16.3	9.6	1.4	-9.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB		-3.7	0.7	5.7	13.2	18.2	22.9	25.5	25.0	21.4	15.0	8.8	1.1	(27)
(28)			MCDB		-2.4	2.5	10.1	21.9	25.9	28.4	29.7	28.3	24.7	19.9	12.5	2.3	(28)
(29)		2%	WB		-7.9	-2.0	3.4	11.3	16.5	21.5	24.0	23.7	19.9	13.0	5.9	-3.1	(29)
(30)			MCDB		-6.7	0.4	7.2	18.4	23.6	26.3	27.8	26.9	23.0	16.5	8.6	-2.1	(30)
(31)		5%	WB		-10.5	-4.6	1.8	9.4	15.2	20.3	23.1	22.7	18.4	11.6	3.6	-6.3	(31)
(32)			MCDB		-9.4	-2.6	5.0	15.3	21.2	24.9	26.7	25.6	21.5	15.3	6.0	-5.1	(32)
(33)		10%	WB		-12.4	-7.2	0.5	7.8	13.9	19.1	22.2	21.8	17.0	10.0	1.5	-9.4	(33)
(34)			MCDB		-11.5	-5.6	3.1	13.0	19.4	23.5	25.5	24.6	20.2	13.5	3.5	-8.3	(34)
(35)	Mean Daily Temperature Range		MDBR		8.9	10.0	9.4	10.1	10.1	9.3	8.0	8.2	10.6	10.6	8.8	8.4	(35)
(36)		5% DB	MCDBR		10.8	11.4	10.7	14.6	13.8	11.8	10.1	9.9	12.4	13.5	10.8	10.3	(36)
(37)			MCWBR		9.7	9.6	6.8	8.2	6.8	5.9	4.7	4.8	6.6	7.9	7.8	9.6	(37)
(38)		5% WB	MCDBR		10.8	11.1	9.8	13.5	12.2	10.3	8.8	8.0	10.1	12.1	10.5	10.4	(38)
(39)			MCWBR		9.8	9.5	6.7	7.9	6.5	5.9	4.7	4.4	6.1	7.9	8.1	9.8	(39)
(40)	Clear-Sky Solar Irradiance	taub			0.310	0.357	0.439	0.550	0.570	0.545	0.508	0.470	0.426	0.449	0.389	0.328	(40)
(41)		taud			2.063	1.919	1.801	1.722	1.749	1.896	2.045	2.146	2.177	1.959	2.014	2.026	(41)
(42)		Ebn at Noon			753	780	763	714	719	742	766	781	780	670	639	671	(42)
(43)		Edh at Noon			99	140	184	219	220	191	163	141	126	135	105	93	(43)
(44)	All-Sky Solar Radiation	RadAvg			2.09	3.23	4.36	4.66	5.22	5.56	5.09	4.66	4.03	2.81	1.91	1.62	(44)
(45)		RadStd			0.11	0.15	0.30	0.48	0.43	0.66	0.49	0.32	0.35	0.21	0.15	0.10	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page